Assignment 1

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1 Camera Calibration

A camera projection matrix $\mathbf{P}$ maps a world point $\mathbf{X_i} = \begin{bmatrix} X_i & Y_i & Z_i & 1 \end{bmatrix}^T$ to its corresponding image point $\mathbf{x_i} = \omega \begin{bmatrix} u_i & v_i & 1 \end{bmatrix}^T$ such that $\mathbf{x_i} = \mathbf{P}\mathbf{X_i}$ which gives two equations with repsect to the unknown parameters of $\mathbf{P} = \begin{bmatrix} m_{ij} \end{bmatrix}$:

$$\begin{aligned} m_{11}X_i + m_{12}Y_i + m_{13}Z_i + m_{14} - m_{31}u_iX_i - m_{32}u_iY_i - m_{33}u_iZ_i - m_{34}u_i = 0 \\ m_{21}X_i + m_{22}Y_i + m_{23}Z_i + m_{24} - m_{31}v_iX_i - m_{32}v_iY_i - m_{33}v_iZ_i - m_{34}v_i = 0 \end{aligned}$$

Given $n \geq 6$ pairs of corresponding points, we have an over-determined system of equations, $\mathbf{Ap} = \mathbf{0}$, where $\mathbf{p}$ is the vectorized form of $\mathbf{P}$.

Consider the SVD of $\mathbf{A}$ as $\mathbf{A} = \mathbf{UDV^T}$. Then the least square solution $\mathbf{p} = \min_p \|\mathbf{Ap}\|$, $\|\mathbf{p}\| = 1$ is the last column of $\mathbf{V}$, which is the eigenvector corresponding to the smallest eigenvalue. And then you can construct the camera projection matrix $\mathbf{P}$ up to scale.

(a) Show that the solution $\mathbf{p}$ is the last column of $\mathbf{V}$ corresponding to the smallest eigenvalue of $\mathbf{A}$.

Theorem 1. *(The singular Value Decomposition, SVD) Let A be an $(m \times n)$ matrix with $m \geq n$. Then there exist orthogonal matrices $\mathbf{U}(m \times n)$ and $\mathbf{V}(n \times n)$ and a diagonal matrix $\mathbf{D} = diag(\sigma_1, ...\sigma_2)(m \times n)$ with $\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_n \geq 0$ such that $\mathbf{A} = \mathbf{UDV^T}$ holds.*

This means that there exists a special orthonormal set of vectors (e.g. the columns of $\mathbf{V}$) that is mapped by the matrix $\mathbf{A}$ into an orthonormal set of vectors (e.g. the columns of $\mathbf{U}$)

Theorem 2. *Let $\mathbf{A} = \mathbf{UDV^T}$. Then the problem*

$$\|\mathbf{Ax}\|_2 = min, \text{ subject to } \|\mathbf{x}\|_2 = 1$$

has the solution $\mathbf{x} = \mathbf{v_n}$ and the value of the minimum is $min_{\|\mathbf{x}\|_2=1} \|\mathbf{Ax}\|_2 = \sigma_n$

Proof. For orthogonal matrix $\mathbf{V}$, let $\mathbf{V^Tx} = \mathbf{y}$: then we have $\|\mathbf{x}\| = \|\mathbf{VV^Tx}\| = \|\mathbf{Vy}\| = \|\mathbf{y}\|$. Thus,

$$\begin{aligned} &\|\mathbf{Ax}\|_2 = min, \text{ subject to } \|\mathbf{x}\|_2 = 1 \\ \Leftrightarrow\ &\|\mathbf{UDV^Tx}\|_2^2 = min, \text{ subject to } \|\mathbf{x}\|_2 = 1 \\ \Leftrightarrow\ &\|\mathbf{UD}(\mathbf{V^Tx})\|_2^2 = min, \text{ subject to } \|\mathbf{VV^Tx}\|_2 = 1 \\ \Leftrightarrow\ &\|\mathbf{UDy}\|_2^2 = \|\mathbf{Dy}\|_2^2 = min, \text{ subject to } \|\mathbf{y}\|_2 = 1 \\ \Leftrightarrow\ &(\sigma_1^2y_1^2 + \sigma_2^2y_2^2 + \cdots + \sigma_n^2y_n^2) = min, \text{ subject to } \|\mathbf{y}\|_2 = 1 \end{aligned}$$

Thus, the problem deforms into a simple equation of finding the minimum value of

$$\sigma_1^2y_1^2 + \sigma_2^2y_2^2 + \cdots + \sigma_n^2y_n^2$$

with respect to $y_1^2 + y_2^2 + \cdots + y_n^2 = 1$. Knowing that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$,

$$\sigma_1^2y_1^2 + \sigma_2^2y_2^2 + \cdots + \sigma_n^2y_n^2 \geq \sigma_n^2 \text{ for } \mathbf{y} = \mathbf{e_n}$$

Finally, we can conclude that our desired $\mathbf{x} = \mathbf{Vy} = \mathbf{v_n}$ □

(b) Given the following corresponding points, determine the camera projection matrix $\mathbf{P}$ using the SVD method.

For the following questions, you can check the [MATLAB] source code in the zip file. Only the results will be given in this report.

Finding $\mathbf{A} \in \mathbf{M}_{40\times 12}$ matrix for $\mathbf{Am} = \mathbf{0}$ with given 20 point informations, and single value decomposing the matrix, we can obtain

$$\mathbf{P} = \begin{bmatrix} 0.0031 & 1.4602e-04 & -4.4850e-04 & -0.9789 \\ 3.0702e-04 & 6.3719e-04 & -0.0028 & -0.2041 \\ 1.6793e-06 & 2.7477e-06 & -6.8396e-07 & -0.0013 \end{bmatrix}$$

(c) Another method to determine the $\mathbf{P}$ is using the pseudo inverse. By setting $m_34 = 1$, we have $\mathbf{Ap} = \mathbf{b}$, and the least square solution $\mathbf{p} = min_{\mathbf{p}}\|\mathbf{Ap} - \mathbf{b}\|$, $\|p\| = 1$ become $\mathbf{p} = \mathbf{A}^{+}\mathbf{b}$, where $\mathbf{A}^{+} = (\mathbf{A^T A})^{-\mathbf{1}}\mathbf{A^T}$ is the pseudo inverse of $\mathbf{A}$. Determine $\mathbf{P}$ for the data in b) using a pseudo inverse method.

We can use the same matrix $\mathbf{A}$ obtained in (b) without the last column (let that matrix be $\mathbf{C} \in \mathbf{M}_{40\times 11}$). Then we should now solve the problem $\mathbf{Cm} = \mathbf{b}$ where $\mathbf{b} \in \mathbb{R}_{40}$ is consisted with image point coordinates.

Using the pseudo-inverse method, we can also obtain

$$\mathbf{P} = \begin{bmatrix} -2.3326 & -0.1100 & 0.3374 & 736.6739 \\ -0.2311 & -0.4795 & 2.0872 & 153.6278 \\ -0.0013 & -0.0021 & 0.0005 & 1.0000 \end{bmatrix}$$

2 Uncalibrated Photometric Stereo

For the following questions, you can check the [Python - Jupyter Notebook] File in the zip file. Only the results will be given in this report.

Apply the uncalibrated photometric stereo method to the given data, and reconstrut possible GBRs. (you may try to use any combination of three or more images in the provided dataset)

(a) Estimate possible surface normal, albedo ($\mathbf{B}$) and light source directions ($\mathbf{s}$) using SVD.

Using SVD, we can estimate one possible solutions of $\mathbf{s}$ and $\mathbf{B}$ (among many) that can regenerate the image given as an input. (Note : I used 5 random images from /CroppedYale/yaleB04 folder.)

For $\mathbf{I_{m\times n}} = \mathbf{U_{m\times m}D_{m\times n}V_{n\times n}} \approx \mathbf{U_{m\times 3}D_{3\times 3}V_{3\times n}}$, we can say that $\mathbf{B} = \mathbf{U_{m\times 3}D_{3\times 3}}$, $\mathbf{S} = \mathbf{V_{3\times n}}$ while choosing the corresponding eigenvectors for the top three greatest eigenvalues in $\mathbf{D}$. The result is given by the following :

```
B =
 [[ -76.73593603 -60.86085945 21.92442088]
 [ -78.27933965 -60.72242006 22.62052301]
 [ -77.79285847 -61.75939217 23.87956924]
 ...
 [-135.80650532 104.01194625 111.81512718]
 [-143.85671177 110.63428455 104.16842392]
 [-143.23867503 110.90525682 102.55368926]]
```

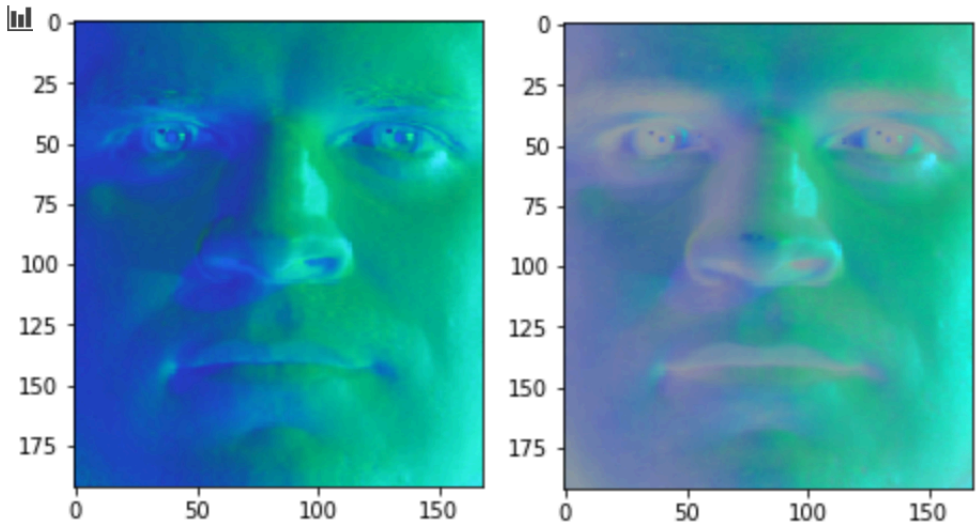

Figure 1: Color-coded image of **Normalized / Unnormalized** Surface Normal Vectors

$\mathbf{S} =$
[[-0.53061609 -0.48704729 -0.10864058 -0.48217144 -0.48676424]
[-0.23304324 -0.63109566 0.37519817 0.60452586 0.20293822]
[0.36320569 0.06960369 0.71317944 -0.03030925 -0.59472127]]

(b) Display the reconstructed surface normals using color code (i.e., map x, y, z components of the surface normal $\mathbf{n}$ linearly in the RGB channels, respectively). Refer to [1]

Left part of Figure 1 is a color-coded version of obtained surface normal vectors, using **normalized surface normals**. Albedo values for each pixels are not embedded in the image.

Right part of Figure 1 is a color-coded version of obtained surface normal vectors, using **unnormalized surface normals**. Albedo values are directly embedded in the image.

Both images are generated by scaled matrices for proper pixel values between 0 and 255.

(c) Synthesize images with some light source direction $\mathbf{s}^*$ which is different from the light sources in $\mathbf{s}$. How does it look?

As shown in Figure 2, I tried several different light source directions to synthesize different images. It does look a bit different from each other, but some images are a bit unnatural.

Also, despite the many attempts of trying different light source directions, similar images were often generated.

(d) Is there any idea to resolve the Bas-Relief ambiguity?

Bas-Relief ambiguity states that, in uncalibrated photometric stereo problems, one cannot accurately specify the real surface normal vectors or albedos (including the light source directions). This is because, exactly same images can be generated for different light source directions and different surface normals in case when the matrix $\mathbf{B}$ and $\mathbf{S}$ are related with $\mathbf{B}^*$ and $\mathbf{S}^*$ via GBR transformation.

To solve this problem, some people gave extra constraints (assumptions) to the problem. One of them is assuming **constant albedo** or **same intensity of all light sources** to specify μ, ν

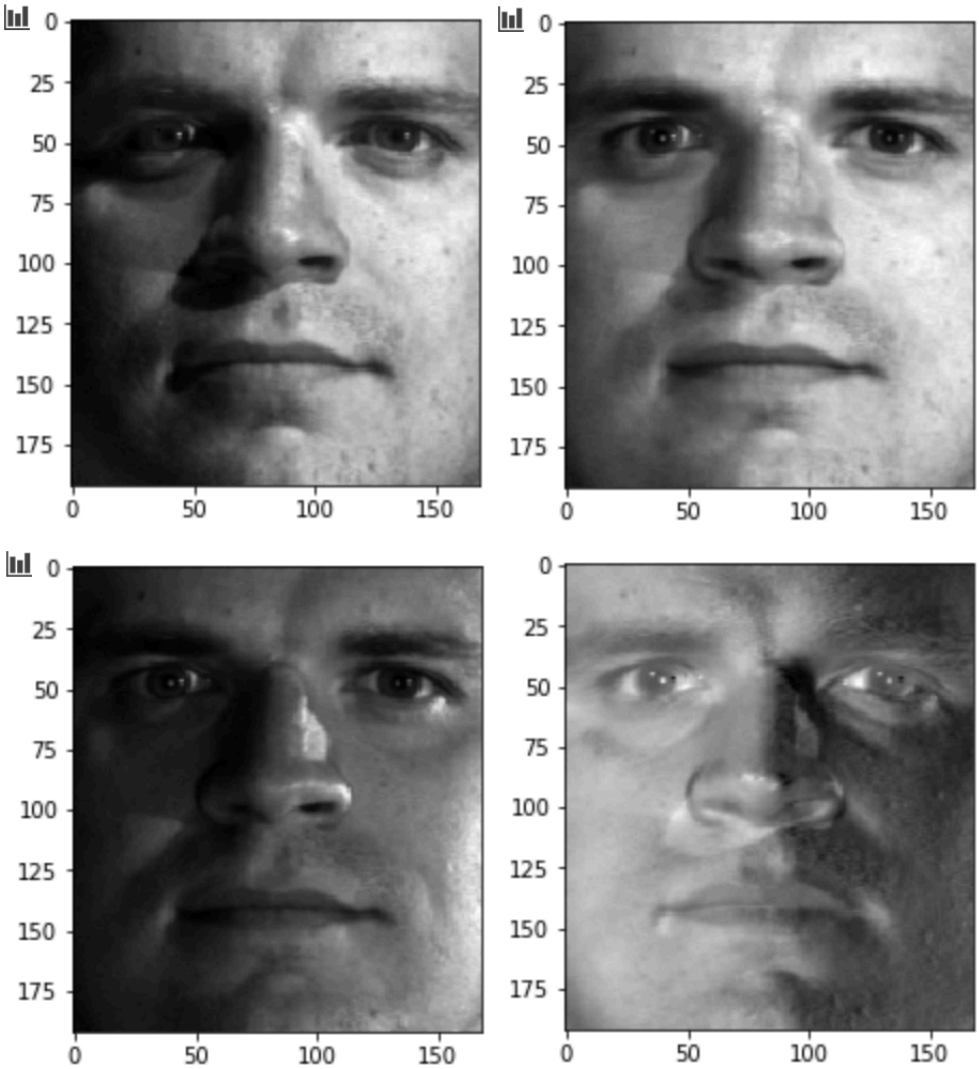

Figure 2: Resynthesized Images with different light source directions
Clockwise direction from top-left image, given light source directions are :

$$\mathbf{s_1} = \begin{bmatrix} -1 & 1 & -1 \end{bmatrix}$$
$$\mathbf{s_2} = \begin{bmatrix} -5 & -1 & 0 \end{bmatrix}$$
$$\mathbf{s_3} = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix}$$
$$\mathbf{s_4} = \begin{bmatrix} -1 & -5 & 5 \end{bmatrix}$$

and λ values necessary for GBR transformation. In fact, the GBR transformation is restricted to the binary subgroup given by $\lambda = \pm 1, \mu = 0$, and $\nu = 0$.

Proof of this is quite simple. If albedo $a(x, y) = |\mathbf{b}(x, y)|$ is constant, this means that $|\mathbf{b}(x, y)| = |\mathbf{b}^*(x, y)| = |\mathbf{A}\mathbf{b}(x, y)|$. This means that matrix $\mathbf{A}$ must preserve vector lengths, which means that $\mathbf{A}$ must be an orthogonal matrix. The only orthonormal matrices with the shape of GBR transformations are actually specified as explained above.

This actually means that if **constant albedo** constraint is given, one can *almost* determine the true surface, up to a sign : $\pm f(x, y)$.